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The Short-Run Policy Constraints of Long-Run Expectations

Contents

1 Overview

The paper argues that monetary policy is constrained not only by conventional frictions (e.g., Phillips-curve tradeoffs or the ZLB), but also by *how private agents form long-run expectations*. If agents extrapolate short-run movements into their perceived long-run "endpoints" for inflation and nominal rates, then aggressive short-run stabilization can destabilize (or strongly distort) long-horizon beliefs. In that case, the central bank cannot implement the same kind of "full control" outcomes that would be feasible under full-information rational expectations (FIRE). The authors build a medium-scale New Keynesian model with a *shifting-endpoints learning mechanism*, estimate it using survey expectations and long-term yields, and quantify welfare and counterfactual policy rules.

2 Roadmap of the paper

Main sections:

1. **Section II:** A stylized NK model to show the *mechanics* of the constraint: extrapolation bias / shifting endpoints can make the inflation target infeasible or require extreme policy paths.
2. **Section III:** A medium-scale NK model with sticky prices and wages, a policy rule, and imperfect-information learning about slow-moving components.
3. **Section IV:** Bayesian estimation using macro data *plus* survey expectations and long-term interest rates; model fit and the inferred learning gain (especially a decline after 1999).
4. **Section V:** Optimal policy under learning (policy aggressiveness is endogenously limited); welfare comparisons and counterfactuals.

3 Section II: A stylized NK model and the policy constraint

3.1 Forward solutions for the IS curve and Phillips curve

Let x_t be the output gap, π_t inflation, R_t the short nominal rate, and r_t^n the natural real rate. Let $\widehat{\mathbb{E}}_t[\cdot]$ denote *subjective* expectations (not necessarily model-consistent). The paper writes the forward-looking solutions (paper eqs. (1)–(2)) as:

$$x_t = \widehat{\mathbb{E}}_t \sum_{T=t}^{\infty} \beta^{T-t} \Big[(1-\beta)\, x_{T+1} - \big(R_T - \pi_{T+1} - r_T^n\big)\Big], \tag{EGP-1}$$

$$\pi_t = \widehat{\mathbb{E}}_t \sum_{T=t}^{\infty} (\xi\beta)^{T-t} \Big[\kappa x_T + (1-\xi)\beta\pi_{T+1}\Big]. \tag{EGP-2}$$

Intuition. When agents extrapolate, the *entire future path* of R_T and π_T embedded in long-horizon expectations feeds back into current x_t and π_t. The monetary authority is therefore interacting with the *expectations formation mechanism*, not just the structural NK block.

3.2 An "ideal" implementation under anchored expectations

In the stylized setup, the paper considers a simple target (e.g., $\pi_t = 0$). The interest-rate path that would implement the target involves long-horizon expectations of future objects (paper eq. (3)):

$$\begin{aligned} R_t = r_t^n &+ \widehat{\mathbb{E}}_t \sum_{T=t}^{\infty} (\xi\beta)^{T-t} \Big[\xi\beta x_{T+1} + (1-\xi)\beta\kappa^{-1}\pi_{T+1} \Big] \\ &+ \widehat{\mathbb{E}}_t \sum_{T=t}^{\infty} \beta^{T-t} \Big[(1-\beta)x_{T+1} - \big(\beta R_{T+1} - \pi_{T+1} - \beta r_{T+1}^n\big) \Big]. \end{aligned} \tag{EGP-3}$$

Under FIRE and well-anchored long-run beliefs, the "endpoint" terms do not drift, and policy can implement the target with a stable instrument path.

3.3 Shifting endpoints for π_t and R_t

The key wedge is a low-frequency component in inflation and the nominal rate:

$$\begin{bmatrix} \pi_t \\ R_t \end{bmatrix} = \bar{\omega}_t + e_t, \qquad \bar{\omega}_{t+1} = \bar{\omega}_t + u_{t+1}, \tag{EGP-4}$$

where $\bar{\omega}_t$ is a persistent "endpoint" (random walk in the stylized section) and e_t is transitory. Agents update their perceived endpoints with a constant-gain rule (steady-state Kalman filter; paper eq. (5)):

$$\omega_{t+1}^{\pi} = \omega_t^{\pi} + \bar{g}\big(\pi_t - \omega_t^{\pi}\big), \qquad \omega_{t+1}^{R} = \omega_t^{R} + \bar{g}\big(R_t - \omega_t^{R}\big). \tag{EGP-5}$$

Interpretation of $\bar{g}$. Larger $\bar{g}$ means agents place more weight on current observations when revising long-run beliefs. So high $\bar{g} \Rightarrow$ "extrapolation" is strong and long-run expectations are *not* well anchored.

3.4 The central constraint: stabilizing inflation moves the endpoint of R

In the stylized model, implementing the target $\pi_t = 0$ requires an interest rate (paper eq. (6)):

$$R_t = r_t^n + \left(\frac{1}{1-\beta} + \frac{(1-\xi)\beta\kappa^{-1}}{1-\xi\beta} \right) \omega_t^{\pi} - \frac{\beta}{1-\beta}\, \omega_t^{R}. \tag{EGP-6}$$

But because ω_t^R is itself updated from R_t, the instrument feeds back into beliefs. Substituting the policy into the learning law yields belief dynamics (paper eq. (7)):

$$\begin{aligned} \omega_{t+1}^{\pi} &= (1-\bar{g})\omega_t^{\pi}, \\ \omega_{t+1}^{R} &= \left(1 - \frac{\bar{g}}{1-\beta}\right) \omega_t^{R} + \bar{g} \left(\frac{1}{1-\beta} + \frac{(1-\xi)\beta\kappa^{-1}}{1-\xi\beta} \right) \omega_t^{\pi} + \bar{g}\, r_t^n. \end{aligned} \tag{EGP-7}$$

The paper's key feasibility condition (paper eq. (8)) is:

$$\bar{g} > 2(1-\beta) \quad \Rightarrow \quad \text{the inflation target cannot be implemented (instrument path destabilizes endpoints).} \tag{EGP-8}$$

Economic intuition of Result 1. If the private sector treats short-run policy moves as evidence about the long-run level of nominal rates, then raising R_t to cool inflation mechanically *raises* ω_t^R. But higher ω_t^R increases future expected nominal rates, pushing up current long rates and undoing part of the intended stabilization. When $\bar{g}$ is large enough, this feedback loop becomes unstable: stabilizing inflation would require ever larger policy moves, which further shift long-run beliefs, and so on.

3.5 Subjective vs objective expectations wedge

A clean summary of the wedge between subjective and objective interest-rate expectations is (paper eq. (9)):

$$\begin{aligned}\widehat{\mathbb{E}}_t R_T - \mathbb{E}_t R_T &= \left(1 - \frac{\phi^{-1} - \beta}{1-\beta}\right)\omega_t^R \\ &= \bar{g}\left(1 - \frac{\phi^{-1} - \beta}{1-\beta}\right)\sum_{j=0}^{\infty}\left(1 - \bar{g}\,\frac{1-\phi^{-1}}{1-\beta}\right)^j r_{t-1-j}^n. \end{aligned} \tag{EGP-9}$$

Interpretation. The wedge is larger when (i) $\bar{g}$ is larger (beliefs react more), and (ii) policy is more "active" (through the policy mapping indexed by ϕ). This is the sense in which the *expectations channel can be a policy constraint.*

4 Section III: Medium-scale NK model with imperfect information and learning

4.1 Policy rule (estimated)

The baseline policy rule in levels (paper eq. (10)) is:

$$R_t = R_{t-1}^{\rho_R}\left[R\left(\frac{P_t}{P_{t-1}}\right)^{1+\phi_\pi}\left(X_t^{CB}\right)^{\phi_x}\right]^{1-\rho_R}\left(\Delta X_t^{CB}\right)^{\phi_{\Delta x}} m_t. \tag{EGP-10}$$

Here X_t is the output gap (ratio of actual to natural output) and X_t^{CB} is the central bank's measured gap (e.g. a real-time gap with measurement error); m_t is a policy shock.

A log-linear approximation highlights the role of a time-varying intercept $\bar{R}_t$:

$$\begin{aligned}\hat{R}_t &= \bar{R}_t + \rho_R \hat{R}_{t-1} + (1-\rho_R)\Big[\phi_\pi \hat{\pi}_t + \phi_x \tfrac{c}{y}(\hat{c}_t - \hat{c}_t^n)\Big] + \phi_{\Delta x}\tfrac{c}{y}\big(\Delta\hat{c}_t - \Delta\hat{c}_t^n\big) + \hat{m}_t, \\ \bar{R}_t &= \tfrac{c}{y}\Big[(1-\rho_R)\phi_x \hat{\eta}_t + \phi_{\Delta x}(\hat{\eta}_t - \hat{\eta}_{t-1})\Big], \end{aligned} \tag{EGP-20}$$

where η_t is the measurement-error component in the perceived gap (so it acts like an "unobserved" policy driver in the data).

4.2 Observation equations: long-term yield and the ZLB shadow rate (empirical measurement block)

Two measurement equations play an important role in identifying the expectations channel.

10-year Treasury yield (expectations hypothesis + time-varying term premium). The observed 10-year yield is modeled as the average of expected future short rates plus a term premium shock:

$$\hat{R}_t^{10yrs} = \bar{R} + \bar{tp} + \frac{1}{40}\,\widehat{\mathbb{E}}_t \sum_{T=t}^{t+40} \hat{R}_T + \hat{tp}_t,$$
$$\hat{tp}_t = \rho_{tp}\hat{tp}_{t-1} + \varepsilon_t^{tp}.$$

Intuition. The 10-year yield helps discipline the model-implied expected path of the short rate. The shock $\hat{tp}_t$ absorbs deviations from the expectations hypothesis (time-varying term premia).

Observed policy rate around the ZLB. During the ZLB period, the short rate in the data is replaced by a shadow rate (Wu–Xia), with a persistent observation error:

$$R_t^{obs} = R_t^{3M\,Tbill} \cdot \mathbf{1}_{\{\text{NoZLB}\}} + R_t^{SW} \cdot \mathbf{1}_{\{\text{ZLB}\}} = \hat{R}_t + \varepsilon_t^{SW} \cdot \mathbf{1}_{\{\text{ZLB}\}},$$
$$\varepsilon_t^{SW} = \rho_{SW}\varepsilon_{t-1}^{SW} + e_t^{SW}.$$

Survey measures of expectations are treated as noisy signals of agents' subjective beliefs (i.i.d. normal observation errors).

4.3 A compact representation of equilibrium dynamics

Let z_t stack the endogenous variables of interest (output gap, inflation, wage inflation, interest rate, etc.). The paper uses a compact representation (paper eq. (11)):

$$z_t = \sum_{s=1}^{3} A_s \widehat{\mathbb{E}}_t \sum_{T=t}^{\infty} \lambda_s^{-(T-t)} z_{T+1} + A_4 z_{t-1} + A_5 \mu_t + A_6 \nu_{t-1} + A_7 u_t. \tag{EGP-11}$$

There are three groups of exogenous processes:

$$\mu_t = \phi_{\mu\mu}\mu_{t-1} + \varepsilon_t^{\mu}, \tag{EGP-12}$$
$$\nu_t = \phi_{\nu\nu}\nu_{t-1} + \varepsilon_t^{\nu}. \tag{EGP-13}$$

Under FIRE, the (linear) equilibrium law of motion implied by the model can be written as (paper eq. (14)):

$$z_t = \phi_{z\nu}\nu_{t-1} + \phi_{zz} z_{t-1} + \phi_{z\mu}\mu_t + \phi_{zu} u_t. \tag{EGP-14}$$

4.4 Perceived law of motion and shifting endpoints in the medium-scale model

Agents are assumed not to observe ν_t and u_t, and to have a misspecified forecasting model with a latent "endpoint" component ω_t (paper eq. (15)):

$$z_t = \omega_t + \phi_{zz} z_{t-1} + \phi_{z\mu}\mu_t + e_t^z,$$
$$\omega_t = \rho_\omega \omega_{t-1} + e_t^{\omega}. \tag{EGP-15}$$

A constant-gain updating rule for the estimate of the endpoint is (paper eq. (16)):

$$\hat{\omega}_{t+1} = \rho_\omega \hat{\omega}_t + \bar{g}\big(z_t - \widehat{\mathbb{E}}_{t-1} z_t\big), \qquad \hat{\omega}_t \equiv \widehat{\mathbb{E}}_{t-1}\omega_t. \tag{EGP-16}$$

A useful long-horizon object is the implied "endpoint" forecast $\bar{z}_t$ (paper eq. (17)):

$$\bar{z}_{t+1} = \rho_\omega \bar{z}_t + \bar{g}(\rho_\omega I - \phi_{zz})^{-1}\big(z_t - \widehat{\mathbb{E}}_{t-1} z_t\big), \qquad \bar{z}_t \equiv (\rho_\omega I - \phi_{zz})^{-1}\hat{\omega}_t. \tag{EGP-17}$$

4.5 Equilibrium under learning: beliefs affect reality

When I evaluate the true evolution of z_t under the agents' subjective expectations, the realized law of motion includes a drift term that depends on beliefs (paper eq. (18)):

$$z_t = \bar{\omega}_t + \phi_{zz} z_{t-1} + \phi_{z\mu}\mu_t + \phi_{zu} u_t, \tag{EGP-18}$$

where the realized drift $\bar{\omega}_t$ combines the endogenous belief component and the unobserved fundamentals. Substituting (EGP-18) into the learning rule yields (paper eq. (19)):

$$\hat{\omega}_{t+1} = \rho_\omega \hat{\omega}_t + \bar{g}(\bar{\omega}_t - \hat{\omega}_t) + \bar{g}\big(\phi_{zu} u_t + \phi_{z\mu}\varepsilon_t^\mu\big). \tag{EGP-19}$$

Interpretation. Temporary shocks can be mistaken for shifts in endpoints, and the resulting belief movements feed back into equilibrium dynamics. This is the core channel that creates (i) persistent wedges between subjective and objective forecasts, and (ii) limits to aggressive stabilization.

5 Section IV: Estimation and empirical implications

5.1 What is being identified by adding surveys and long rates?

A central identification idea is: *long-horizon survey expectations and long-term yields contain information about the agents' perceived endpoints.* With only standard macro data, a low-frequency latent component can be hard to distinguish from persistent structural shocks. Surveys help pin down $\bar{z}_t$ (the endpoints), and long rates discipline expected paths of short rates.

5.2 Table 1: posterior parameters

Table 1 reports priors and posteriors for structural, policy, learning, and shock-persistence parameters. The big parameter for the expectations channel is the *learning gain*:

- Pre-1999: $10 \times \bar{g} \approx 0.59 \Rightarrow \bar{g} \approx 0.059$.
- Post-1999: $10 \times \bar{g}_{1999} \approx 0.20 \Rightarrow \bar{g}_{1999} \approx 0.020$.
- The endpoint persistence is very high: $\rho_\omega \approx 0.994$.

Interpretation. Since about 1999, long-run expectations appear to become *more anchored* (lower $\bar{g}$), so policy moves translate less into endpoint revisions.

5.3 Table 2: shock volatilities and the Great Moderation

Table 2 reports posterior shock standard deviations separately for 1964–1983 vs. 1984–2021. Many innovation variances shrink in the later subsample (e.g. observed macro shocks and several unobserved shocks), consistent with Great Moderation-style volatility reduction. This matters because a smaller shock environment also makes it easier for agents to learn (and for policymakers to stabilize) without destabilizing endpoints.

TABLE 1
PRIOR AND POSTERIOR PARAMETERS DISTRIBUTION

	Prior			Posterior		
	Distribution	Mean	5%–95%	Mode	Mean	5%–9%
Structural:						
$100 \times (\beta^{-1} - 1)$	G	.15	.03, .34	.13	.14	.10, .18
σ^{-1}	G	.55	.10, 1.62	.11	.11	.10, .13
b	B	.70	.43, .92	.76	.77	.75, .78
ι_w	B	.40	.01, .91	.74	.74	.71, .78
ξ_p	B	.60	.34, .83	.83	.81	.79, .84
Monetary policy:						
ϕ_π	G	.50	.13, 1.07	.21	.22	.12, .34
ρ_R	B	.75	.46, .95	.93	.93	.92, .93
ϕ_x	G	.10	.01, .28	.09	.09	.08, .10
$\phi_{\Delta x}$	G	.15	.04, .32	.07	.07	.07, .08
Learning:						
$10 \times \bar{g}$	G	.353	.036, .947	.600	.587	.547, .628
$10 \times \bar{g}_{1999}$	G	.353	.036, .945	.193	.197	.173, .222
ρ_ω	B	.949	.881, .989	.994	.994	.993, .995
Exogenous (unobserved):						
ρ_ζ	B	.95	.88, .99	.99	.98	.98, .99
ρ_η	B	.95	.88, .99	.98	.98	.97, .99
Exogenous (observed):						
ρ_g	B	.60	.25, .90	.89	.89	.87, .90
ρ_a	B	.60	.25, .90	.79	.77	.75, .80
ρ_φ	B	.60	.25, .90	.63	.62	.60, .65

NOTE.—The posterior distribution is obtained using a sequential Monte Carlo algorithm. B = beta; G = gamma.

TABLE 2
PRIOR AND POSTERIOR DISTRIBUTION: SHOCK INNOVATIONS

	Prior			Posterior 1964–83		Posterior 1984–2021	
	Distribution	Mean	5%–95%	Mean	5%–95%	Mean	5%–95%
Unobserved:							
σ_{θ_p}	IG	.10	.02, .28	.04	.02, .07	.04	.02, .07
σ_{θ_w}	IG	.10	.02, .28	.17	.14, .20	.08	.07, .09
σ_m	IG	.10	.02, .28	.13	.12, .15	.07	.06, .08
σ_ζ	IG	.10	.02, .28	.16	.13, .18	.16	.13, .18
σ_η	IG	1.00	.32, 2.45	1.19	1.07, 1.30	.71	.66, .77
Observed:							
σ_g	IG	1.00	.32, 2.45	.93	.84, 1.04	.52	.47, .57
σ_a	IG	.10	.02, .28	.05	.04, .06	.02	.02, .03
σ_φ	IG	.10	.02, .28	.04	.03, .04	.02	.02, .02

NOTE.—The posterior distribution is obtained using a sequential Monte Carlo algorithm. IG = inverse gamma.

5.4 Table 3: model comparison across specifications

Table 3 compares marginal data densities (MDD) and implied learning parameters across variants. My reading:

- Allowing a post-1999 gain change is strongly supported (baseline fits much better than "single gain").
- Rational expectations (FIRE) fits much worse than learning with shifting endpoints.
- The estimated persistence ρ_ω is very stable across specifications.

Core implication: the data prefer a model where long-run expectations adjust over time, with a sizable decline in the adjustment speed after 1999.

TABLE 3
ALTERNATIVE MODEL SPECIFICATIONS

	MDD (1)	$\bar{g}$ (2)	$\bar{g}_{1984}$ (3)	$\bar{g}_{1999}$ (4)	ρ_ω (5)
Baseline	1,000	.055–.063		.017–.022	.993–.995
Gain in 1984	1,002	.062–.073	.040–.050	.016–.021	.993–.995
Single gain	967	.039–.044			.993–.995
Constant volume	946	.049–.057		.015–.019	.994–.996
FIRE	893				
No ZLB sample		.049–.058		.009–.017	.993–.995

NOTE.—Column 1 shows the marginal data density (MDD) of each model, with the exception of the specification with a shorter sample. Columns 2–4 show the 5–95 percentile range estimates in the parameters of agents' updating rule.

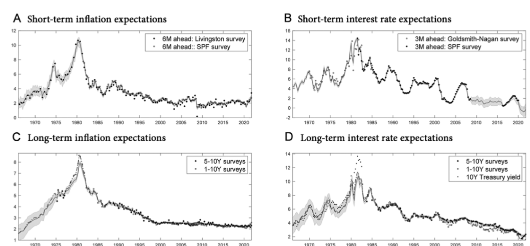

FIG. 1.—Fitting survey expectations. The four panels show model predictions together with survey measures. *A*, *B*, Evolution of short-term expectations. Dark gray lines denote median model predictions, and light gray areas mark the 95% coverage interval. Filled squares and filled circles show survey data. *C*, *D*, Long-run forecasts. Dark gray (dot-dashed black) lines show model predictions for the 5- to 10-year-ahead (1- to 10-year-ahead) expectations. Similarly to *A* and *B*, the light gray area measures the 95% coverage interval for 5- to 10-year-ahead predicted forecasts, while filled squares and open circles indicate survey data.

5.5 Figure 1: fitting survey expectations

Figure 1 shows the model matches both short-horizon and long-horizon survey expectations for inflation and interest rates. This is important because it validates the mechanism: the estimated $\bar{z}_t$ and the gain process produce realistic joint dynamics of macro variables *and* expectations.

5.6 Figure 2: monetary transmission depends on the learning gain

Figure 2 reports impulse responses to an expansionary monetary policy shock. Under learning, a key extra object is the response of *long-horizon interest-rate expectations* (dashed line in the figure).

A higher gain implies larger and more persistent movements in long-run expectations, which changes the strength and persistence of output/inflation responses. Lower post-1999 gain yields smaller movements in the long-run expectation component, consistent with better anchoring.

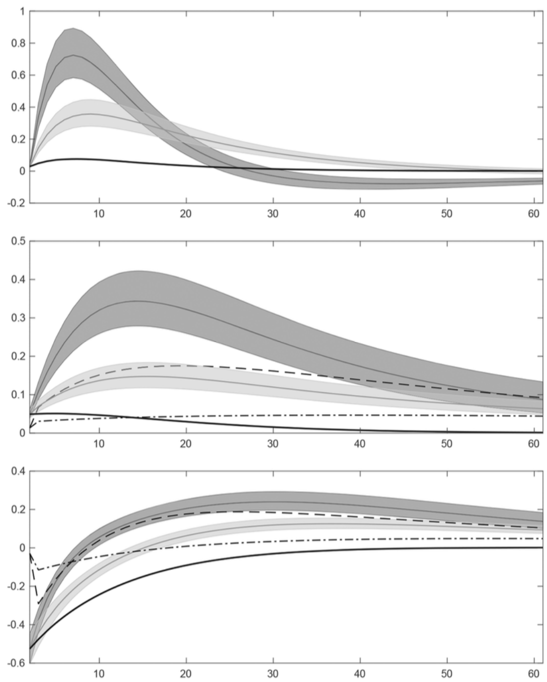

FIG. 2.—Monetary transmission mechanism. The three panels show the impulse responses of output, inflation, and short-term interest rate to a 1% expansionary monetary policy shock under the gains $\bar{g}$ and $\bar{g}_{1999}$. The dark (light) gray solid lines denote median responses for output, inflation, and interest rate under $\bar{g}$ ($\bar{g}_{1999}$) and the dark (light) gray area measures the 95% coverage interval. Dashed ($\bar{g}$) and dash-dotted ($\bar{g}_{1999}$) lines measure 5- to 10-year-ahead expectations. Finally, the solid black lines show the model under rational expectations.

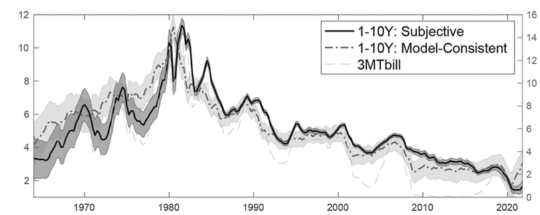

FIG. 3.—Subjective versus objective expectations. This figure shows model predictions for 1- to 10-year-ahead interest rate expectations. The solid black line shows median predictions for agent's subjective expectations, while the dot-dashed gray line measures median objective (model-consistent) expectations. The dark gray and light gray areas capture the 95% coverage interval. Finally, the dashed light gray line (and the right scale) shows the short-term interest rate.

5.7 Figure 3: subjective vs. objective long-horizon expectations

Figure 3 compares subjective interest-rate expectations (what agents forecast under their perceived model) to objective model-consistent expectations. A persistent wedge is the empirical signature of extrapolation bias / shifting endpoints: agents can think "the long run moved" when fundamentals did not, and vice versa. This wedge is exactly the object that creates a policy constraint in Section II and in the full model.

6 Section V: Optimal policy under learning

6.1 Welfare criterion

The policymaker chooses a sequence of policy instruments to minimize an intertemporal quadratic loss (paper eq. (21)):

$$\mathcal{L} = \mathbb{E}_t \sum_{T=t}^{\infty} \beta^{T-t} L_T. \tag{EGP-21}$$

The key subtlety is that L_T depends on outcomes that are influenced by *belief dynamics*, so policy affects welfare partly through how it shapes long-run expectations.

6.2 Targeting rule under "full control" (a useful benchmark)

If the policymaker had full control of aggregate demand, the Ramsey problem implies a target criterion (paper eq. (22)):

$$0 = \bar{\phi}_\pi \pi_t + \left(\pi_t^w - \iota_w \pi_{t-1}\right) + \bar{\phi}_{\Delta x}\left(x_t^{CB} - x_{t-1}^{CB}\right). \tag{EGP-22}$$

A natural policy implementation is a partial-adjustment rule (paper eq. (23)):

$$R_t = R_{t-1} + \phi\Big[\bar{\phi}_\pi \pi_t + \left(\pi_t^w - \iota_w \pi_{t-1}\right) + \bar{\phi}_{\Delta x}\left(x_t^{CB} - x_{t-1}^{CB}\right)\Big], \tag{EGP-23}$$

where ϕ governs aggressiveness: $\phi \to \infty$ corresponds to implementing the target criterion "as if" the central bank had full control.

TABLE 4
OPTIMAL POLICY

	1964–83		1984–99		2000–2021	
Baseline versus:	Median	5%–95%	Median	5%–95%	Median	5%–95%
Historical policy	.68	.48–.82	.53	.36–.67	.60	.48–.70
Full control of aggregate demand	1.61	1.48–1.77	1.71	1.60–1.85	1.55	1.47–1.65
FIRE	1.63	1.51–1.81	1.75	1.63–1.90	1.57	1.48–1.68
Small cost push	2.80	2.21–3.61	3.16	2.61–3.85	2.57	2.13–3.15
Optimal ϕ:						
Prior	661	2.94–∞	661	2.94–∞	661	2.94–∞
Posterior	.44	.24–.84	.78	.55–1.12	6.77	4.44–10.1

NOTE.—The top section of the table shows the relative welfare loss under baseline optimal policy, as compared with alternative specifications. The results report optimal policy outcomes under the parameters' posterior distribution. The bottom section shows the optimal response coefficient ϕ in our baseline optimal policy under the prior and posterior parameters' distribution, respectively.

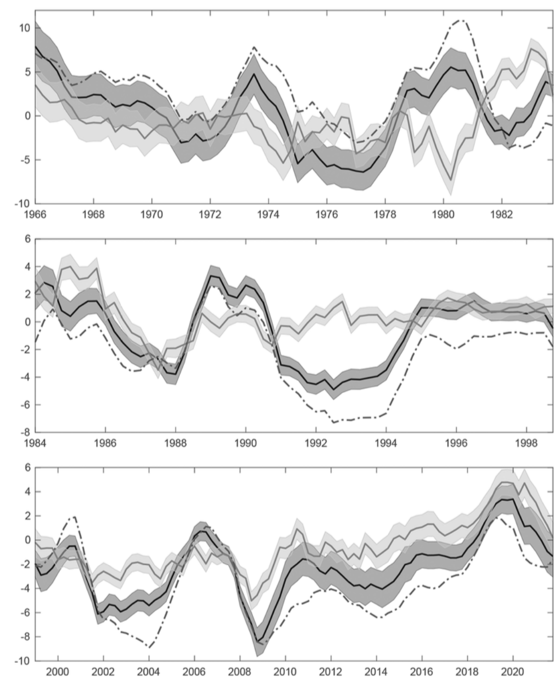

FIG. 4.—Output gap counterfactuals. The three panels show the evolution of the output gap under different policy regimes. The black solid line and dark gray area show median predictions and the 68% coverage interval for the baseline optimal policy specification. The gray solid line and light gray area show median predictions and the 68% coverage interval for the optimal policy under full aggregate demand control. Finally, the dashed dark gray line measures the median prediction of the output gap under the historical policy.

6.3 Table 4: welfare comparisons and the optimal aggressiveness parameter

Table 4 compares baseline optimal policy to alternative environments/policies. My interpretation:

- **Baseline vs. historical policy:** the baseline optimal policy substantially lowers losses (ratios below 1).
- **Baseline vs. full control:** the full-control benchmark dominates (ratios above 1), especially post-2000.
- **Optimal ϕ over time:** with posterior parameters, optimal policy is more gradual pre-1984, less gradual in 1984–1999, and very aggressive in 2000–2021 (consistent with the post-1999 lower gain / better anchoring).

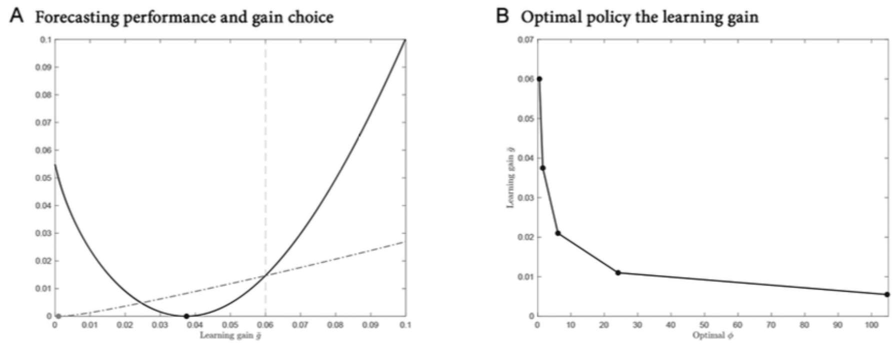

FIG. 5.—Endogenous learning gain. *A*, RMSE as a function of the learning gain under two policies: the baseline optimal policy (black line) and the optimal policy under full control of aggregate demand (dark gray dash-dotted line). For clarity, for each policy specification, the RMSEs are expressed in percentage deviations from their minimum value (depicted by the filled circles). The light gray dashed line corresponds to the modal estimate of the learning gain. *B*, Optimal policy coefficient ϕ corresponding to different learning gains. Starting from the estimated gain, $\bar{g} = 0.06$, each subsequent gain minimizes the RMSE, conditional of the optimal policy being in place. As the "optimal" gain is reduced, optimal policy becomes more aggressive.

6.4 Figure 4: output-gap counterfactuals

Figure 4 plots counterfactual output gaps under: (i) historical policy, (ii) baseline optimal policy under learning, (iii) the full-control benchmark. The main message is that optimal policy improves stabilization, but the gap between baseline and full control remains nontrivial, illustrating that endogenously-moving long-run expectations limit what the interest-rate instrument can achieve.

6.5 Figure 5: endogenous learning gain and policy feedback

Figure 5 studies how forecast performance (RMSE) varies with the learning gain under different policies. It highlights an equilibrium feedback:

1. policy affects macro volatility and persistence,
2. that changes the forecasting environment for agents,
3. which changes the gain that minimizes forecast errors,
4. which then changes the tightness of the policy constraint.

The paper formalizes a "bounded rationality" selection for the gain (paper eq. (24)):

$$\mathbb{E}[\mathrm{RMSE}(\bar{g})] \leq \min_{\tilde{g}} \mathbb{E}[\mathrm{RMSE}(\tilde{g})] + \epsilon. \tag{EGP-24}$$

Intuition. More stable policy can encourage smaller gains (more anchoring), relaxing the constraint further, but the gain is ultimately pinned down by the forecasting problem agents face (not by the policymaker directly).

7 Short summary

The paper's punchline is a synthesis:

- Long-run expectations are an endogenous state variable.
- If agents extrapolate short-run movements into long-run endpoints, then aggressive short-run stabilization can be counterproductive or infeasible.
- In US data, the learning gain falls markedly after about 1999, consistent with improved anchoring of expectations.
- Optimal policy becomes more aggressive when expectations are anchored, but in earlier periods gradualism is optimal because it avoids destabilizing endpoints.

8 Conclusion

8.1 Summary

Eusepi, Giannoni, and Preston (2026) argue that monetary policy is constrained not only by familiar frictions (e.g., Phillips-curve tradeoffs, the ZLB, state uncertainty), but also by a distinct and quantitatively important *information friction*: the **stability of long-run interest-rate expectations**. When private agents form long-horizon beliefs using *shifting endpoints* (i.e., they update perceived long-run drifts/endpoints using short-run forecast errors), movements in the short policy rate can induce **excessive revisions in long-term expectations**. This breaks the tight mapping from the policy rate to long-term interest rates that is typically assumed under full-information rational expectations (FIRE), *even if the expectations hypothesis is taken seriously as an accounting identity for yields.* As a result, **aggressive short-run stabilization can be counterproductive** because it generates **undesirable volatility in long-term rates and aggregate demand** through standard intertemporal-substitution channels. Therefore, the central bank **optimally chooses a less aggressive policy** than would be recommended by a FIRE analysis, at least in regimes where long-run expectations are relatively unanchored.

8.2 Main conclusions

1. **A new policy constraint: imperfect control of the term structure of expectations.** If long-run expectations respond endogenously and excessively to short-rate movements, then policy cannot freely use large short-rate adjustments without destabilizing long-term rates.
2. **Price stability can be desirable but not feasible in the short run.** Relative to many imperfect-information models where full stabilization is feasible but not optimal, here the paper emphasizes a stronger claim: stabilizing inflation and activity as in textbook NK/FIRE analyses may be *desirable* but *infeasible* when long-run expectations are fragile.
3. **Optimal policy exhibits "gradualism" when expectations are unanchored.** When the economy is characterized by a high learning gain (agents update endpoints quickly), the optimal rule is substantially less aggressive toward inflation/output gaps than under FIRE.
4. **Anchoring long-run expectations expands the feasible set of stabilization policies.** Once long-run expectations are stabilized, more aggressive business-cycle stabilization becomes feasible, so the constraint is fundamentally about the *short run* versus the *long run.*
5. **Quantitatively important in U.S. postwar data.** The estimated model implies the constraint is not a small theoretical curiosity: it materially changes optimal policy design and welfare.

8.3 Key quantitative messages (to cite in discussion)

Two empirical results summarized in the paper are especially useful for interpretation:

- **Estimated learning gain declines after 1999 (expectations become more anchored).** The paper estimates a learning gain of about 0.06 before 1999 and about 0.02 after 1999. Interpreting the gain: a 1% forecast error implies roughly a 6 basis point change in the perceived long-run drift before 1999, versus about 2 basis points after 1999. This pins down *how strongly short-run surprises feed into long-run beliefs.*

- **Welfare comparisons and the "limits of monetary policy" (Table 4).** Baseline optimal policy (under learning) improves substantially on the historical policy rule: the baseline loss is roughly 30–50% lower than the historical rule depending on the subsample. However, counterfactual environments where the central bank has "full control" (or where expectations are FIRE) deliver markedly better outcomes than the learning baseline, highlighting that *belief dynamics materially constrain what policy can achieve.* Consistently, the estimated optimal aggressiveness parameter is much smaller in earlier subsamples (when expectations are less anchored) and becomes very large in the 2000–2021 period (when the gain is low and the constraint relaxes).

8.4 Economic intuition (why the constraint exists)

The constraint is easiest to understand as an expectations-amplification loop:

1. **In standard NK/FIRE logic,** the central bank influences long-term real rates mainly by controlling expected future short rates, so aggressive short-rate policy can quickly stabilize inflation and activity.

2. **With shifting-endpoints learning,** agents partially interpret short-run interest-rate movements as evidence about long-run "endpoints" (long-run inflation and nominal-rate drifts). Hence, a policy-induced rise in short rates can trigger *excessive upward revisions in long-run nominal-rate expectations.*

3. **Long-term rates then become excessively sensitive.** Because the long rate embeds long-horizon expectations, the long rate can move too much (and too persistently) relative to the policy move.

4. **Aggregate demand reacts to long-term rates.** Excess volatility in long-term real rates creates excess volatility in consumption/investment via intertemporal substitution, making aggressive stabilization *welfare-worsening.*

5. **Therefore, optimal policy is constrained in the short run.** The central bank chooses a more inertial/gradual approach, aiming to *avoid destabilizing long-run expectations.* Only as long-run beliefs become more anchored (lower gain) can the central bank safely adopt more aggressive stabilization policies.

8.5 Takeaway

When long-run interest-rate expectations are formed through shifting endpoints that overreact to short-run policy moves, the central bank has imperfect control of the term structure, so optimal monetary policy must be less aggressive in the short run—and anchoring long-run expectations is a prerequisite for effective business-cycle stabilization.